CHANDU PRASAD BHAIRAPU



GenAI Developer

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ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in LLM applications grounded in company data, from retrieval-augmented generation, embedding pipelines and tool use to evaluation and API design in Python. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, backed by RAG evaluation research at Covestro.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present — Muldestausee, Germany

- Owned the knowledge-extraction pipeline end-to-end as sole developer, from data ingestion and chunking to prompt engineering and indexing (agentic, deterministic and monolithic strategies), reporting directly to the CTO and working with SAP domain experts.
- Designed the agent architecture in LangGraph (entity-extraction, relationship-analysis, consolidation and reflection agents in parallel) for complex regulatory and technical documents; evaluation against an SAP-system-derived ground truth raised precision/recall from 33%/45% to 93%/90%.
- Engineered a deterministic, code-based extraction pipeline for a second, high-volume source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing accuracy toward 100%.
- Designed and implemented a Model Context Protocol (MCP) service that exposes the knowledge graph for millisecond-latency retrieval, integrated into Claude Code for agentic development workflows.
- Developed Expert Interface, a graph curation web app (React frontend, Python microservices backend) on Azure used by domain experts to validate and approve knowledge-graph changes.
- **Tech stack: Python, LangGraph, LangChain, LiteLLM, MCP, RAG, GraphRAG, Vector Search, Neo4j, MongoDB, Postgres, FastAPI, Docker, Kubernetes, Azure, React**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 — Leverkusen, Germany

- Designed and deployed an agentic AI application with LangGraph that uses tool use to route requests across vector database queries, general inference and summarization, after working with business stakeholders on their decision-making workflows.
- Built the Python/FastAPI backend (API design, Postman tests) and a React frontend, containerized with Docker and deployed to AWS with CDK; versioned in GitLab with CI/CD pipelines.
- **Tech stack: Python, LangGraph, LangChain, FastAPI, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD, React**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 — Leverkusen, Germany

- Automated evaluation of retrieval-augmented generation for the PrismaChat RAG system: generation and retrieval metrics with RAGAS and PytrecEval against chemist-provided ground truth, on serverless, queue-based asynchronous AWS infrastructure (Lambda, SQS, DynamoDB) built with CDK.
- Benchmarked multilingual embedding models on over 1M chunks of internal chemical data and designed RAG pipelines: chunking strategies, vector databases, indexing and fine-tuning models for retrieval.
- **Tech stack: Python, RAGAS, PytrecEval, LangChain, Hugging Face Transformers, PyTorch, AWS (Lambda, SQS, DynamoDB, CDK), Azure AI Search, Docker**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost and LSTM models to predict prepolymer synthesis outcomes from time-series process data, and delivered a Streamlit dashboard for chemists to simulate parameter changes before running physical batches.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, NumPy, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XGBoost regressor, FB Prophet and LSTM models for EV load time-series forecasting; the LSTM reached 96% test accuracy with strong performance on unseen data.

PROJECTS

Project: Price Intelligence Copilot | Python | FastAPI | pgvector | TypeScript | MCP

- Conversational analyst over commodity price data: an agent with typed function calling pulls exact figures from a price-index database and explains moves with cited passages from market reports retrieved by hybrid search (pgvector + keyword), with a grounding check on every answer.
- Async embedding pipeline and FastAPI services for Q&A, semantic search and price alerts; MCP server in TypeScript (Node.js) for enterprise access; labelled eval set comparing two foundation models on quality, cost and latency in CI.

TECHNICAL SKILLSET

- **LLM Applications:** RAG (chunking, retrieval strategies, vector databases, grounding), evaluation (RAGAS, PytrecEval, ground truth), function calling and tool use, MCP, agent architectures (LangGraph, LangChain), prompt engineering, LiteLLM (provider-agnostic access to foundation models), GraphRAG
- **Backend:** Python, FastAPI, API design, Python microservices, SQL, data modelling (Neo4j, Postgres, MongoDB), React (beginner)
- **Search & Data:** vector databases (Qdrant, Pinecone, Azure AI Search), embedding model benchmarking, Hugging Face Transformers, PyTorch, Pandas
- **Cloud & DevOps:** Docker, Kubernetes, Azure, AWS (Lambda, SQS, DynamoDB, S3, CDK), Git, GitLab CI/CD; AI-assisted development with Claude Code

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering — August 2017 – July 2021
Minor in Computer Science & Engineering
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS & CERTIFICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles
- Udemy - LLMs Mastery : Transformers & Generative AI

LANGUAGES

English (Business Fluent) | German (B1) | Hindi (Fluent) | Telugu (Native)